

Yu Li

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Education

George Washington University Washington, D.C.
Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0 Aug 2025 – May 2029
 Advisor: [Prof. Tian Lan](#) 📧; Mentor: [Prof. Zhengling Qi](#) 📧 (expected)

Wuhan University, Hongyi Honor College Wuhan, China
B.Eng. in Microelectronics Science and Technology, GPA: 3.87/4.0 Sept 2021 – Jun 2025
 Advisor: [Prof. Cheng Lei](#) 📧 & [Prof. Wei Liu](#) 📧

Research Interests

My research focuses on **LLM post-training**, **agent policy learning**, and **generative AI**. Specific topics include LoRA fine-tuning, preference optimization, reasoning, agentic RL, and multimodal in-context learning.

Preprints & Under Review

- [P.1] [OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning](#) 📧
 Yu Li, Rui Miao, Tian Lan, Zhengling Qi
Under review
- [P.2] [ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning](#) 📧
 Yu Li, Rui Miao, Zhengling Qi, Tian Lan
Under review
- [P.3] **MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference**
 Yu Li, Binxu Li, Tian Lan
Under review
- [P.4] [Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO](#) 📧
 Yu Li, Tian Lan, Zhengling Qi
Under review
- [P.5] [INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization](#) 📧
 Yu Li, Tian Lan, Zhengling Qi
Under review & AI with Recursive Self-Improvement Workshop at ICLR, 2026

Publications

C=Conference, J=Journal, †=Equal Contribution

- [C.1] **CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion**
 Yu Li, Yujun Cai, Chi Zhang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- [C.2] **Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization**

Yu Li, Sizhe Tang, Tian Lan

Findings of the Association for Computational Linguistics (ACL), 2026

[C.3] KG-SAM: Injecting Anatomical Knowledge into Segment Anything Models via Conditional Random Fields

Yu Li, Da Chang, Xi Xiao

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026 [Oral]

[C.4] ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense

Yu Li[†], Sizhe Tang[†], Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan

Intelligent Cloud Computing and Networking Workshop at INFOCOM, 2026

[C.5] Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT

Da Chang, Peng Xue, Yu Li, Yongxiang Liu, Pengxiang Xu, Shixun Zhang

The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026

[C.6] Mixed Text Mode Optical Character Recognition Based On Transformer

Yu Li[†], Da Chang[†]

International Conference on Neural Information Processing (ICONIP), 2024

[C.7] ECG Classification with Dual Models: XGBoost Voting and Deep Learning with Attention

Yu Li, Yunhao Hu, Jie Chen, Binghao Wang, Wei Liu *et al.*

International Conference on Advanced Computer Technology and Electronics, 2023

[J.1] Dual branch SAM-Transformer Fusion Network for Accurate Breast Ultrasound Image Segmentation

Yu Li, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

Medical Physics, 2025

[J.2] SfMDiffusion: Self-supervised Monocular Depth Estimation in Endoscopy Based on Diffusion Models

Yu Li, Da Chang, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

International Journal of Computer Assisted Radiology and Surgery, 2025

[J.3] Local Optimum Time-Reassigned Synchrosqueezing Transform for Bearing Fault Diagnosis

Site Lv, Shan Zeng, Yu Li, Ke Yang, Yulong Chen

IEEE Sensors Journal, 2024

Selected Awards & Honors

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- **ICML 2026 Gold Reviewer**, top reviewers recognition. 2026
 - **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. 2024
 - **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. 2023, 2024
 - **First-Class Scholarship**, Top 5%, ¥3,000, three times. 2022, 2023, 2024

Academic Services

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- **Conference Reviewer:** {COLM, ICML, CVPR, ICLR, AAAI, ICASSP}'26
 - **Journal Reviewer:** Neurocomputing, IEEE Transactions on Networking

Skills

- **Languages:** English (TOEFL 105), Chinese (Native), Japanese (Basic)
- **ML/LLM Stack:** PyTorch, DeepSpeed, Flash Attention, vLLM, OpenRLHF, VERL, TRL
- **Tools & Platforms:** Linux/Ubuntu, Docker, Git, Weights & Biases, Cadence